ADNOC GAS OPERATIONS & MARKETING LLC

PUNCHLIST MANAGEMENT PROCEDURE

EPCm CONTRACT NO.: 3000076501
ADNOC GAS PROJECT NO.: 30201-50150-LT-HE

LONG TERM - EXECUTE STAGE

PROJECT 5.0 (P5)

EPCm FOR HABSHAN DBN

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TABLE OF CONTENTS

1.0 INTRODUCTION 4
1.1 BACKGROUND 4
1.2 LONG TERM EPCM PROJECT OBJECTIVES 4
1.3 SCOPE BATTERY LIMIT 5
2.0 PURPOSE OF THIS DOCUMENT 5
3.0 ORDER OF DOCUMENT PRECEDENCE 5
4.0 REFERENCE, CODES AND STANDARDS 7
4.1 REFERENCE DOCUMENTS 7
4.2 ADNOC STANDARDS 7
4.3 INTERNATIONAL CODES AND STANDARDS 7
5.0 DEFINITIONS 7
5.1 GENERAL DEFINITIONS 7
5.2 TECHNICAL DEFINITIONS 8
5.3 ABBREVIATIONS 11
6.0 SCOPE 13
7.0 ROLES AND RESPONSIBILITIES 13
7.1 COMPLETION TEAM 13
7.2 CONSTRUCTION/FABRICATION TEAM 14
7.3 COMMISSIONING TEAM 14
7.4 CLIENT/OPERATIONS TEAM 15
7.5 VENDORS 15
8.0 PUNCHLIST MANAGEMENT 15
APPENDIX-01 GREENFIELD & BROWNFIELD PUNCHLIST RAISE FORM 32

LIST OF TABLES

Table 4-1: Reference Documents 7
Table 4-2: ADNOC Standards 7
Table 4-3: International Codes and Standards 7
Table 5-1: General Definitions 7
Table 5-2:Technical Definitions 8

HOLD LIST

PARAGRAPHS HOLD IN THIS ISSUE			
Hold No.	Section / Para No.	Page No.	Holds Description

1.0 INTRODUCTION

1.1 BACKGROUND

ADNOC aims to increase oil production to produce 5.0 MMBOPD by year-end 2027. Accordingly, associated gas production is anticipated to increase. Also, to meet the future oil and gas demand, non-associated gas developments are considered. Hence, the overall gas supply scenarios for major ADNOC Gas (hereafter referred to as COMPANY) assets i.e., Das Facilities, Bab and Habshan Complex (Habshan 0, 1, 2, 4, and 5), Asab and Bu Hasa, is anticipated to change depending on their feed sources, profiles including rates, compositions, and operating conditions. These changes will also subsequently impact the COMPANY downstream facilities, particularly, products handling and transferring facilities i.e., Sales Gas Network, NGL Pipeline Network, Ruwais Fractionation Trains (1, 2, 3, 4 and upcoming new Train 5 part of Bab Cap Gas SELECT Study), Sulphur Handling Facilities, and Condensate systems, etc.

COMPANY has carried out the Long Term DEFINE for the project with M/s Amec International Ltd. (M/s Wood) and subsequently decided to proceed with Long Term EXECUTE phase execution to ensure the readiness of COMPANY facilities to accommodate increased gas supply by December 2027.

Accordingly, COMPANY has awarded EPCm contract to M/s Amec International Ltd. (M/s Wood) to carry out the Long Term scope requirements. M/s Wood shall act as a single entity in the performance of the services and a single point of accountability to ADNOC Gas for the successful completion of Long Term EXECUTE phase.

1.2 LONG TERM EPCM PROJECT OBJECTIVES

The main objective of the Long Term EPCm is to execute the Habshan DBN Project to support ADNOC to achieve oil production target of 5.0 million BOPD so that Associated and Non-Associated Gas can be processed within the overall processing capacity of Habshan. The Long Term EPCm scope includes the following major areas as a minimum:

1. Habshan Complex (H0, H1 Train 1/2, H2, H4, H3 for SS-300 Tie-in)
2. Bab CDS and Bab NGL
3. Habshan 5
4. Pipeline scope activities including below new pipelines
   I. Bab CDS to Habshan Complex HP pipeline
   II. Bab CDS to Habshan Complex MP pipelines (2 no.s)
   III. Habshan 5 to Habshan 2 pipeline

The scope also includes adequacy checks for Ruwais NGL Fractionation Trains, MERAM,

Habshan Sulphur Granulation Plant, Sales gas, NGL, Condensate and interconnecting Pipeline network and Utilities for all impacted AG facilities.

The Long Term EPCM scope includes the following major activities as a minimum:

1. Detailed Engineering and design services
2. Procurement Services
3. Construction Management services
4. Pre-commissioning, Commissioning, Start-up services
5. Final documentation handover and Project close-out

Project also involves major interface activities with other on-going projects. Refer Project Interface management plan (Doc Num: PP-30201_50150_LT_HE-GG-000-50-00-006) for complete list of interface projects.

1.3 SCOPE BATTERY LIMIT

Refer Design Basis (Doc Num: BOD-30201_50150_LT_HE-GG-000-00-00-001) for the scope battery limits, Gas allocation and detailed Design basis.

2.0 PURPOSE OF THIS DOCUMENT

This document is designed to provide clear and comprehensive guidelines outlining the recommended methodology for the implementation, recording, clearing and accepting of punch lists and punch list items on P5 HABSHAN EPCM project.

3.0 ORDER OF DOCUMENT PRECEDENCE

In the event of any conflict between the applicable documents, the order of precedence shall be as follows:

Note-1 AGES Addendum/AGES shall be considered wherever AGES addendum/AGES is available. DGS Addendum/DGS shall be considered only in case AGES Addendum/AGES is not available.

In the event of any conflict between the applicable documents, the order of precedence shall be as follows: the ANNEXURES or among the ANNEXURES, the order of precedence below shall prevail and shall guide the construction of any part of the AGREEMENT:

1. (a) this FORM OF AGREEMENT;
2. (b) ANNEXURE 1 – SPECIAL CONDITIONS;
3. (c) ANNEXURE 2 – GENERAL TERMS AND CONDITIONS;
4. (d) ANNEXURE 11 – COMPLETION AND ACCEPTANCE REQUIREMENTS

5. (e) ANNEXURE 15 – RESPONSIBILITIES MATRIX;
6. (f) ANNEXURE 13 – ICV IMPROVEMENT PLAN;
7. (g) ANNEXURE 12 – HSE REQUIREMENTS;
8. (h) ANNEXURE 3 – SCOPE OF WORK;
9. (i) ANNEXURE 7 – INSURANCE;
10. (j) ANNEXURE 10 - PROJECT EXECUTION REQUIREMENTS; k) ANNEXURE 14 - QUALITY CONTROL AND INTEGRITY MANAGEMENT (QA/QC);
11. (l) ANNEXURE 16 - KPI SCHEDULE;
12. (m) ANNEXURE 5 - PRICING SCHEDULE;
13. (n) ANNEXURE 17 - TOTAL INSTALLED COST;
14. (o) ANNEXURE 4 - COMPLETION SCHEDULE;
15. (p) ANNEXURE 9 - CONTRACTOR' S SUBMISSIONS.
16. (q) ANNEXURE 6 - COMPANY PROVIDED FACILITIES;
17. (r) ANNEXURE 8 - SECURITY AND FORMS;
18. (s) ANNEXURE 18 - FORM OF EPCM SUPPLY AGREEMENT; and
19. (t) ANNEXURE 19 - FORM OF EPCM CONSTRUCTION CONTRACT.

However, in case of conflict between the various specifications, standards and the requisition, most stringent requirements shall govern. EPCm CONTRACTOR/VENDOR shall refer any such conflict, and any other matter, which requires further interpretation, to the attention of COMPANY and COMPANY's interpretation shall be final. Resolution of any conflict shall be obtained from COMPANY in writing before proceeding.

Any deviation from COMPANY standard accepted in the existing facility design shall not be considered as basis for this PROJECT design and any deviations shall be submitted to COMPANY for approval before proceeding.

4.0 REFERENCE, CODES AND STANDARDS

4.1 REFERENCE DOCUMENTS

Table 4-1: Reference Documents

Sl. No.	Document Number	Document Title
1.		

4.2 ADNOC STANDARDS

Table 4-2: ADNOC Standards

Sl. No.	Document Number	Document Title
1.		

4.3 INTERNATIONAL CODES AND STANDARDS

Table 4-3: International Codes and Standards

Sl. No.	Document Number	Document Title
1.		

5.0 DEFINITIONS

5.1 GENERAL DEFINITIONS

For this document, the following general definitions are applicable.

Table 5-1: General Definitions

DEFINITION	DESCRIPTION
COMPANY	ADNOC GAS (AG) for whom the SERVICES are being performed by the EPCm CONTRACTOR.
EPCm CONTRACTOR	Amec International Limited – Abu Dhabi (WOOD) appointed by COMPANY to perform Engineering, Procurement, Construction management, Pre-Commissioning and Commissioning stage for Long Term Solution (under EXECUTE phase).
FACILITIES	Habshan Complex (H0, H1 Train 1/2, H2, H4, H3 for SS-300 Tie-in), Bab CDS, Bab NGL, Habshan 5 and interconnecting pipelines.

DEFINITION	DESCRIPTION
LICENSOR	Supplier of packaged units and responsible for the operating and design performance.
MANUFACTURER/ VENDOR/ SUPPLIER	The Party (parties) which manufactures and/or supplies material, equipment, technical documents or drawings and/or services to perform the duties specified by the EPCm CONATRCATOR
PMC	Project Management Consultant appointed by COMPANY for the PROJECT
PMT	Project Management Team
PROJECT	PROJECT 5.0 (P5), Long Term EPCm for HABSHAN DBN
SERVICES	The scope of work to be completed during the EXECUTE stage (EPCm) of the PROJECT.
SHALL	The use of the word “shall” indicate a mandatory requirement
SHOULD	The use of the word “should” indicate a strong recommendation to comply with the requirements of this document
EPCm CONSTRUCTION CONTRACTOR	Subcontractor appointed by EPCm CONTRACTOR to carry out Construction works at site for respective construction packages.
SUB-CONTRACTOR	The COMPANY approved party, which carries out services subcontracted by EPCm CONTRACTOR.

5.2 TECHNICAL DEFINITIONS

For this document, the following technical definitions are applicable.

Table 5-2:Technical Definitions

DEFINITION	DESCRIPTION
Acid Gas	A gas stream containing predominantly H_2S and/or CO_2 from an Acid Gas Removal Unit
Condensate	A hydrocarbon liquid stream containing Pentane+ compounds
Dehydrated Gas	A gas stream that has been dehydrated to remove water

DEFINITION	DESCRIPTION
Dry Gas	A gas stream containing no liquids at flowing conditions, but which may still contain water and hydrocarbons that will fall out if there is a change in flowing conditions.
Feed Gas - Associated Gas	A Gas stream that is produced from an oil well during the extraction of crude oil.
Feed Gas – Non-associated Gas	Hydrocarbon gas originating from gas reservoirs.
GMP 2023	Gas Master Plan 2023
Lean Gas	Lean nature of gas is defined based C2+ in the gas stream. If C2+ is less than 15% then stream is defined as Lean gas
Mixed Feed Gas Stream	Mixed associated and non-associated gas stream
NGL	A liquid stream with primarily C2, C3, C4 and some C5+ compounds.
Rich Gas	Rich gas is defined based C2+ in the gas stream. If C2+ is more than 15% it is considered rich gas
Sales Gas	Residue gas sent to the Sales gas network.
Sour Gas	A gas stream containing significant amounts of H_2S and/or CO_2.
Sustainable Profile (GMP 2023)	GMP 2023 High Supply Scenario – unconstrained oil production from GMP Guidelines Document from all associated gas fields
Sweet Gas	A gas stream in which the H_2S and CO_2 are not present in significant amounts.
Technical Profile	AON provided profile for technical case for associated gas.
Technical Optimized case	The ADNOC Gas facilities (Das, Asab-0/3, Buhasa, Ruwais, associated pipelines) will be sized to accommodate technical rates from associated gas fields.

DEFINITION	DESCRIPTION
	Habshan facilities will be sized to accommodate technical rate from all associated gas fields (Bab, Buhasa, NEB, Offshore) but not simultaneously. All inlet facilities / inlet compression shall be designed for technical rate.
Unconstrained Production Profile / High Supply Scenario	This shall be based on Upstream's OPEC Unconstrained Oil production case. This scenario corresponds to sustainable projects profiles
1 Tier Punchlist Method	With the 1 tier punchlist the punchlist items are input to a default project punchlist and are linked to a system and subsystem. This is the recommended methodology on Greenfield projects.
2 Tier Punchlist Method	With the 2 tier punchlist method a punchlist is raised for each discipline and MC1, SH1 or IRN at the time the certification is created in the CMS. The punchlist items are then linked to it when the walkdown is performed upon completion of construction or commissioning for the given scope. This is the recommended methodology on Brownfield projects.
Brownfield	Brownfield refers to the modification/upgrade to an existing facility. In GoTechnology® philosophy, the completion of Brownfield construction and commissioning activities is recorded on certificates which reference inspection and testing procedures.
Certification and Commissioning Handover Pack (CCHP)	Compilation of all completions documentation required to demonstrate that technical integrity has been achieved for a specific part of a project scope of work. CCHP's are usually compiled by the project completions contractor.
Construction Workpack	Compilation of all engineering documentation required to fabricate, assemble, inspect and test a defined part of a project scope of work. Construction workpacks are usually compiled by the project construction contractor.
Greenfield	Greenfield refers to the construction of a new asset (platform, module, unit, facility). In GoTechnology® philosophy, the completion of Greenfield construction

DEFINITION	DESCRIPTION
	and pre-commissioning activities is recorded on Inspection and Test Records (ITRs).
MC1	Mechanical completion handover certificate (Brownfield).
Punchlist	List of defects or deviations from the original design.
Punchlist Item	An individual defect or deviation from the original design which has been identified, recognised and recorded during the process of construction, testing or start-up. Punchlist items must be reported and tracked to ensure their criticality is assessed and their clearance planned in accordance with the handover process.
SH1	System handover (Brownfield).

5.3 ABBREVIATIONS

ADNOC	Abu Dhabi National Oil Company
AGP	ADNOC Gas Processing
AGES	ADNOC Group Engineering Specifications
CRS	Comment Resolution Sheet
CSU	Commissioning and Start-Up
D&A	Decommissioning & Abandonment
DGS	Design General specification
DEFINE	Front End Engineering Design (FEED)
EPCM	Engineering Procurement & Construction Management
FEED	Front End Engineering Design
IFA	Issued For Approval
IFR	Issued For Review

CCHP	System handover (Brownfield)
CMS	Completions Management System
CWP	Construction Work Pack
DAC	Discipline Acceptance Certificate
FAT	Factory Acceptance Test
FCO	Facilities Coordinator Officer
IRN	Inspection Release Note
ITR	Inspection and Test Record
MRO	Material Requisition Order
OIM	Offshore Installation Manager
PL	Punch List

6.0 SCOPE

This procedure provides the necessary steps required to manage punchlists and punchlist items throughout the P5 HABSHAN EPCM project lifecycle. The methodology follows best practice and describes the process with due consideration to all project phases from fabrication through construction, pre-commissioning and commissioning to plant start-up.

This document is applicable to Greenfield, Brownfield and decommissioning projects and to any Completions Management Systems (CMS).

The instructions provided in the current document must be read in conjunction with the following documents:

- COM-PLN-100001 Corporate Completions Delivery Plan
- COM-PRO-100050 Recommended Practice for the Population of a GoTechnology Brownfield Project
- COM-PRO-100051 Recommended Practice for the Population of a GoTechnology Greenfield Project
- WCS-OPS-PHL-400 GoTechnology®
- WCS-OPS-PHL-402 Greenfield Handover Process
- WCS-OPS-PHL-403 Decommissioning Activity Flowchart
- WCS-OPS-FLC-410 Functional Certification Flowchart - Greenfield
- WCS-OPS-FLC-411 Functional Certification Flowchart - Brownfield
- WCS-OPS-FLC-412 Functional Certification Flowchart – Decommissioning

7.0 ROLES AND RESPONSIBILITIES

7.1 COMPLETION TEAM

The Completions/Certification team are responsible for:

- Producing a project specific Punchlist Management Procedure
- Specifying the minimum data requirements for all punchlists and punchlist items. These should be communicated to all project personnel involved in the punchlist process
- Ensuring all punchlists and punchlist items recorded by any party meet the data requirements communicated to the project
- Rejecting punchlist items which do not meet the specified requirements, advising and assisting wherever possible to resolve missing or required data
- Creating punchlists(s) in CMS (Brownfield)

- Inputting all punchlist items in CMS (Greenfield and Brownfield) and verifying accuracy of data
- Creating and maintaining the Master Punchlist Book (Greenfield)
- Updating the status of punchlist items in CMS when these have been cleared by the 'Actionee' (groups responsible for actioning punchlist items)
- Updating the status of punchlist items in CMS when these have been accepted by the 'Responsible person' (groups responsible for accepting punchlist items)
- Running punchlist reports from CMS
- Arranging tri-party walkdown inspections

7.2 CONSTRUCTION/FABRICATION TEAM

The Construction/Fabrication team are responsible for:

- Raising and agreeing punchlist items
- Providing the Completions team with required data for input to the CMS
- Nominating persons, generally from each relevant discipline, to take part in tri-party or discipline acceptance walkdowns
- Attending tri-party walkdowns
- Actioning punchlist items which have been assigned to the Construction group
- Clearing and accepting punchlist items and updating the Completions team
- Agreeing transfer of punchlist item ownership, where applicable, at handover stages

7.3 COMMISSIONING TEAM

The Commissioning team are responsible for:

- Raising and agreeing punchlist items
- Providing the Completions team with required data for input to the CMS
- Nominating persons, generally from each relevant discipline, to take part in tri-party or discipline acceptance walkdowns
- Attending tri-party walkdowns
- Actioning punchlist items which have been assigned to the Commissioning group
- Clearing and accepting punchlist items and updating the Completions team
- Agreeing transfer of punchlist item ownership, where applicable, at handover stages

7.4 CLIENT/OPERATIONS TEAM

The client's Operation team are responsible for:

- Raising and agreeing punchlist items
- Providing the Completions team with required data for input to the CMS
- Nominating persons, generally from each relevant discipline, to take part in tri-party or discipline acceptance walkdowns
- Attending tri-party walkdowns
- Actioning punchlist items which have been assigned to the Operations group
- Clearing and accepting punchlist items and updating the Completions team
- Agreeing transfer of punchlist item ownership, where applicable, at handover stages

7.5 VENDORS

The Vendor's are responsible for:

- Complying with the project Punchlist Management Plan
- Raising and agreeing punchlist items at the time of the inspection
- Providing the Completions team with required data for input to the CMS (via procurement)
- Clearing punchlist items (either at workshop or on site) and updating the Completions team

8.0 PUNCHLIST MANAGEMENT

Document authors should define the key inputs and outputs (i.e. materials, services, information or data that is used as a feed into the document and the deliverables that are produced as a result of this documents application) of the process/procedure/guideline as an introduction to this section

Different approaches have been developed which dictate how punchlists are managed on both Greenfield and Brownfield projects. While the 1 tier method is recommended on Greenfield project, Brownfield utilise the 2 tier method.

8.1 1 Tier vs 2 Tier Punchlist Methods

8.1.1 1 Tier Method

This method requires the Completions team to utilise the project default master punchlist. All punchlist items are added to this list and managed therein. The punchlist items are also linked

to a system and sub system.

8.1.2 2 Tier Method

This method requires the Completions team to create a punchlist per discipline and handover (MC1/SH1) or IRN. All punchlist items are then added to the appropriate list and managed therein.

8.2 Greenfield Punchlist Requirements

The following section describes the punchlist lifecycle and reporting thereof throughout a Greenfield project.

8.2.1 Principle of Punchlists

The 1 tier method is widely regarded as the best approach for punchlist implementation on Greenfield projects. This does not require the creation of a punchlist in the first instance therefore when using the 1 tier method only punchlist items are input to the CMS.

A Punchlist Raise Form is used for recording punchlist items. This form should be added to the pre-commissioning packs such as piping test packs, mechanical packs and loop test packs. Any other discipline packs containing pre-commissioning ITR's should also include this template.

The template is for recording punchlist items only and does not allow for items to be cleared or accepted. Under no circumstances should the Completions team accept signatures on these templates for any reason other than inputting items to the CMS.

8.2.2 Punchlist Preparation Phase

Following the guidelines in this document a project specific procedure should be written and approved by the client prior to project execution. A Project Signature Matrix should also be created at this stage to record all personnel authorised to raise, clear and accept punchlist items.

8.2.2.1 Authorised Persons

All personnel involved in the punchlist process must be identified and listed on the project signatory matrix.

The following information should be populated into the CMS:

- Name
- Company
- Position

8.2.2.2 Punchlist Type

Punchlist types refer to the workgroups involved in the project punchlist process.

The following punchlist types are typical on most Greenfield projects and should be populated into the CMS:

- Construction
- Commissioning
- Fabrication Contractor
- Vendor
- Client/Operations
- Engineering

8.2.2.3 Punchlist Party

Parties are responsible for inspecting (raising), correcting (clearing) and accepting punchlist items.

The following parties are typical on most Greenfield projects and should be populated into the CMS:

- Construction Contractor(s)
- Commissioning Contractor(s)
- Fabrication Contractor(s)
- Vendors
- Client(s)
- Engineering Contractor(s)

This functionality allows the relevant contractor/company to be allocated responsibility of punchlist items.

8.2.2.4 Punchlist Categories

Greenfield projects typically use three punchlist categories A and B however this can vary depending on the client process. In the CMS each item is allocated the appropriate category. The rules governing category allocation are as follows:

Category A

Items to be repaired, completed or corrected before equipment can be handed over from the Construction group to the Commissioning group. By definition all category 'A' items should be cleared prior to acceptance by Commissioning (at the end of construction or pre-commissioning depending on the contractual agreement).

Category B

Items to be repaired, completed or corrected before the equipment can be handed over from the Commissioning group to the Client Operations/Project group. By definition all category 'B' items should be cleared prior to acceptance at the RSFU handover gate.

Items to be repaired, completed or corrected that will not endanger personnel, plant or

equipment. By definition all category 'B' items should be cleared or agreed upon before the end of the project.

8.2.3 Raising Punchlists Items

The personnel raising the punchlist item(s) should use the Punchlist Raise Form template, completing as a minimum, all mandatory data required for input by the Completions team. All Punchlist Raise Forms should be returned to the completions office and placed in the punchlist in-tray.

Blank Punchlist Raise Form templates should always be available in the completions office.

Punchlist items can be either raised in the construction stage inspections, the pre-commissioning packs when carrying out the inspection and testing activities or at the time of the tri-party walkdowns before the handover is submitted.

8.2.3.1 Tri-Party Walkdowns

Tri-party walkdowns are arranged by the Punchlist Coordinator. An invitation to attend the walkdown shall be sent two working days prior to the walkdown date. This will be sent to a pre-defined list of competent representatives from the parties involved in the handover as well as the client.

Should a party representative not be able to attend the walkdown, a competent substitute should be nominated, and the Punchlist Coordinator informed. Walkdowns shall not be cancelled or rescheduled due to the absence of a party.

The walkdowns are usually performed by discipline. On the day of the walkdown, the Punchlist Coordinator shall provide the inspection party with a copy of the system/subsystem mark-ups showing the limits of the subsystem to be inspected along with a punchlist raise form to raise any punchlist item(s). All items must be raised on the aforementioned form. The parties attending the walkdown shall agree the validity of the punchlist items, their categorisation as well as the party responsible for clearing each item. All representatives attending the walkdown shall sign the punchlist raise form, confirming their attendance and the acceptance of each item being raised.

Upon completion of the walkdown, in the event that no defects or non-conformance are identified, the text 'NO PUNCHLIST ITEMS' should be clearly marked on the punchlist form. All parties in attendance should sign the form to indicate their agreement prior to the handover taking place.

The punchlist raise form should then be returned to the Completions office.

Finally, upon completion of the walkdown, assuming there are no blocking punchlist items outstanding, the Completions team should generate the Discipline Acceptance Certificate for signing. This should be signed by the personnel attending the walkdown and passed back to the Completions team for input into the CMS.

Refer to COM-PRO-100011 - System Handover and Completions Dossier Compilation for

further information on handover requirements.

8.2.4 Adding Punchlist Items and Punchlist Data Requirements

It is normally the responsibility of the Punchlist Coordinator to input punchlist items into the CMS. The Punchlist Coordinator will input items from all Punchlist Raise Forms placed in the punchlist in-tray into the CMS on a daily basis.

The Punchlist Coordinator is responsible for requesting clarification from the relevant punchlist parties if the data provided on the punchlist raise form is insufficient.

8.2.4.1 Punchlist Item – Minimum Data

Punchlist Item – An incremental number

Description – An accurate description of the non-conformance or defect to be rectified or verified. Details should include, wherever possible, tag number and relevant drawings. A detailed description of the location should also be included in order that the defective equipment can be found on site

Action Discipline – The discipline who will action the punchlist item

Category – Either A or B

Raised By – A person from the Project Signature Matrix with sufficient permission to raise punchlist items

Date Raised – The date the punchlist item was recorded

System – The affected system

Sub System – The affected subsystem

Action by – The designated party responsible for rectifying the non-conformance or defect (clearance body). In the CMS this is usually recorded in the ‘Cleared By’ and ‘Cleared date’ fields

Tag Number – The tag number of the defective equipment

8.2.4.2 Punchlist Item – Additional Data

ITR/Cert – If the punchlist item relates specifically to an ITR for any given equipment then the ITR should be included where possible

Location – Selected from the list of locations within the CMS

Manhours – An estimate of the manhours required to complete the punchlist item

Responsibility – The designated party responsible for verifying that the defect or non-conformance has been rectified (acceptance body). In the CMS this is usually recorded in the ‘Accepted By’ and ‘Accepted date’ fields

Materials – The MRO number, if material is on order

All punchlist raise forms, when not kept in a pre-commissioning pack, should be filed in a

binder by date order

8.2.5 Managing the Punchlist Items

Using the 1 tier method, as each punchlist item is entered into the CMS, it is allocated a sequential number from PL000001 onwards. In order that all punchlists entered into the CMS can be viewed and prepared for clearance and acceptance they should be printed and filed in binders within the completions office thereby creating a 'Master Punchlist Book'. By doing so the punchlist items are available to any personnel involved in the process to view and sign off at all times. This is particularly useful when punchlist items require update or review as it does not require involvement of the Completions team.

From the date the first punchlist item is added into the CMS, the punchlist items raised over the last week, as required or at a frequency to be determined by the project should be printed out. The report used for the Master Punchlist should have both cleared by/cleared by date and accepted by/accepted by date fields included. This allows for the master copy to be maintained and for those involved in clearing or accepting items to visit the Completions office and sign off items which have been actioned.

8.2.6 Punchlist Clearance and Acceptance

The Punchlist Coordinator should export the outstanding punchlist items by discipline, area, and action by, for distribution to the parties involved in the punchlist clearance process. This should be as required or at a frequency to be determined by the project.

The punchlist parties will use this report to identify their scope of work. Under no circumstances should this report be used to sign off punchlist items. All clearance and acceptance must be recorded on the master punchlist book.

Clearing and accepting punchlist items shall be carried out by competent personnel usually from the work group indicated by the Punchlist Party assignment.

Once the personnel in charge of clearing the punchlist item is satisfied that the defect has been rectified, they should locate the relevant item(s) within the master punchlist book and clear the item(s) by signing and dating the cleared by and cleared date fields.

The personnel responsible for accepting the punchlist item should verify that the defect or non-conformance has been rectified. Following verification, they should locate the relevant item(s) within the master punchlist book and accept the items by signing and dating the accepted by and accepted date fields.

As some signatures are not always easily identifiable it is good practice to add a full name, signature and date. Stickers or other method of indicating which items/sheets have been signed should be used in order that the Punchlist Coordinator can identify those for input into the CMS. The Punchlist Coordinator will input this data into the CMS on a daily basis. At the time of input, the Punchlist Coordinator shall identify (tick) the item clearance and acceptance date fields in the master punchlist book. This will identify the data which has not been input to the CMS.

The Punchlist Coordinator is not authorised to sign off any punchlist items using their own account unless the items are either related to completions documentation or to the CMS.

8.2.7 Punchlist Reports

Three types of punchlist reports are typically required during the execution of the work:

- Report for the master punchlist book:
  - Used to print the master punchlist book on which to record wet signatures for clearance and approval by punch list parties before data is input to the CMS
  - Lists all punchlist items sorted in numerical order
  - Can be filtered by date recorded (in order to print only the punchlist items raised since the last printing)
  - No other filter shall be applied (subsystem, discipline or other) in order to keep the punchlist items in numerical order
  - Includes cleared by and accepted by name and date fields for personnel to sign
- Punchlist status and progress report to project management:
  - Used to report the progress of punch listing by discipline, area, phase, etc to the project management team
  - Is not a list of punchlist items
  - Shows summary counts of punchlist items by disciplines, category and any other relevant breakdown
  - Shows outstanding and closed items
  - Shows a progress curve by discipline of items raised and closed over time
  - Punchlist report provided to all punchlist parties, by discipline, for punchlist clearance:
  - Used to provide a list of outstanding items that need to be actioned by the relevant punch list parties
  - Lists all outstanding items relevant to the discipline/company/any other relevant filter
  - Does not have sign off boxes
  - Is sorted in numerical order
- Punchlist status report for inclusion in handover dossiers
  - Used to report the outstanding and closed punchlist items at the time of handover
  - List of punchlist items relevant to the system/subsystem, sorted in numerical order
  - Includes cleared by and accepted by names and dates
  - Could be the same report used for the Master Punchlist Book

The Punchlist Coordinator will produce these reports and distribute to the required personnel on a regular basis, the frequency of which (weekly, bi-weekly, daily) should be determined

by the project requirements.

8.2.8 Vendor Punchlist

All vendors must comply with the project punchlist procedure which should be issued through procurement.

Any issues found during FAT's must be recorded on a punchlist form. This should either be the punchlist raise form used by the project or on the vendor equivalent. Vendor punchlist forms should only be used if the data requirement matches the project specification.

Upon completion of the FAT, the punchlist must be reviewed and its content and actionees approved by the client/EPC representative in attendance.

Once transmitted to the project this form must be passed to the Completions team for input to the CMS. The copy of the vendor punchlist must be filed with the punchlist raise forms.

8.2.9 Punchlist Process Flow

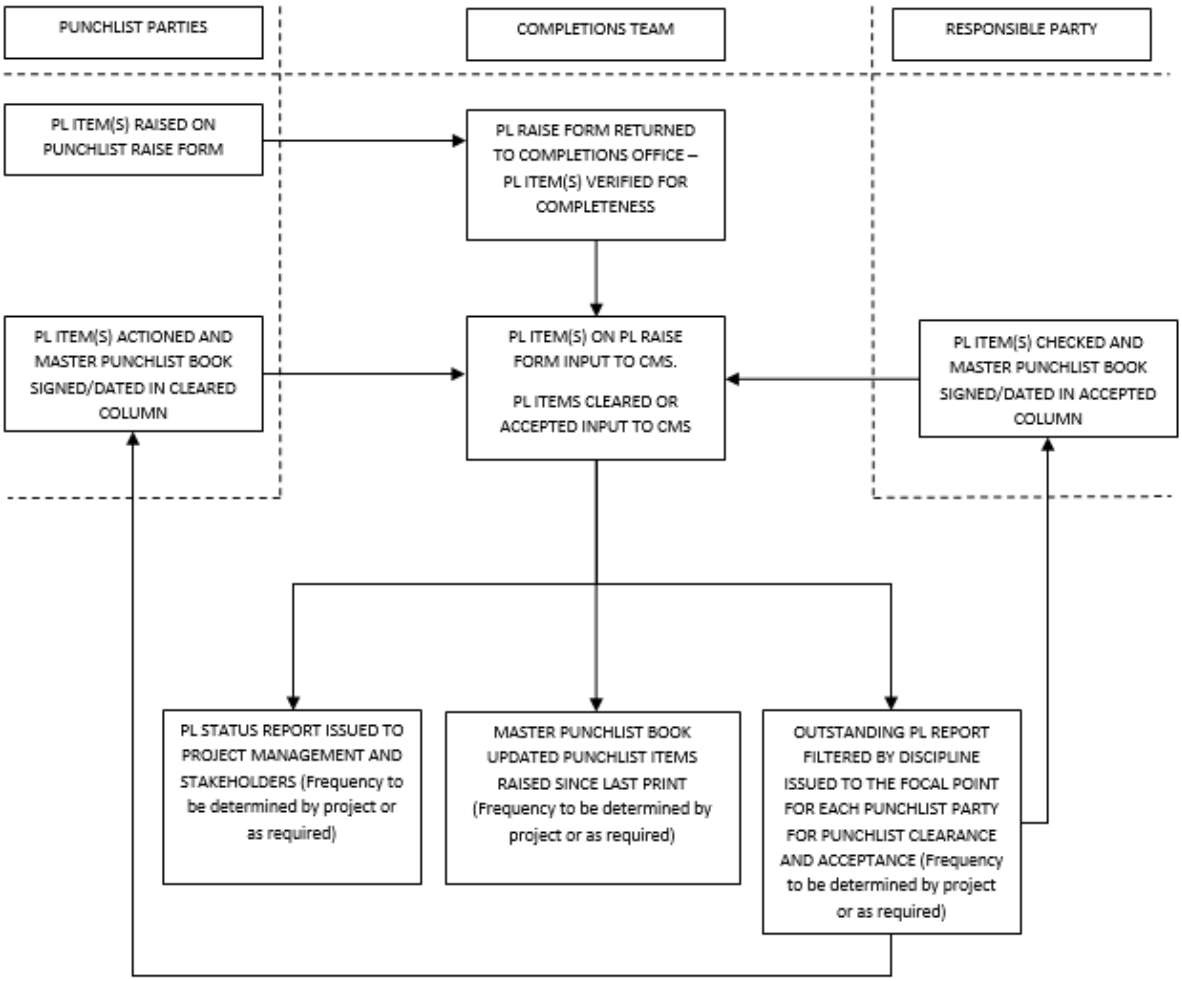

Figure 1 – Greenfield punchlist process flowchart

8.3 Brownfield Punchlist Requirements

The following section describes the punchlist lifecycle and reporting thereof throughout a Brownfield project.

8.3.1 Principle of Punchlists

The 2 tier method is widely regarded as the best approach for punchlist implementation on Brownfield projects.

The main difference between the 1 tier method used on Greenfield and the 2 tier method implemented on Brownfield is that punchlists must be manually created in the CMS, normally on a per discipline basis, for each MC1 and SH1 handover. This is done by the Certification Engineer when all other certification is added to the construction workpack prior to the pack being approved for construction and sent to site.

A punchlist is also required, by discipline, for each piece of equipment being supplied by a vendor. A punchlist form should be transferred with all documentation from the vendor/manufacturer to the project at the time the IRN is approved. The punchlist form is added to the relevant construction workpack and is used for clearance and acceptance of items.

Punchlist items are recorded on the punchlist form before they are added in the CMS.

Note that this principle implies that each MC1 or SH1 scope of work is covered in a single construction workpack and is not shared across several workpacks.

8.3.2 Punchlist Preparation Phase

Following the guidelines in this document a project specific procedure should be written and approved by the client prior to project execution. A Project Signature Matrix should also be created at this stage to record all personnel authorised to raise, clear and accept punchlist items.

8.3.2.1 Authorised Persons

All personnel involved in the punchlist process must be identified and listed on the project signatory matrix. The following information should be populated into the CMS:

- Name
- Company
- Position

8.3.2.2 Punchlist Type

Punchlist types refer to the workgroups involved in the project punchlist process.

The following punchlist types are typical on most Brownfield projects and should be populated into the CMS:

- Construction
- Commissioning
- Fabrication Contractor
- Vendor
- Client/Operations
- Engineering

This functionality allows the relevant contractor/company to be allocated responsibility of punchlist items.

8.3.2.3 Punchlist Party

Parties are responsible for inspecting (raising), correcting (clearing) and accepting punchlist items.

The following parties are typical on most Brownfield projects and should be populated into the CMS:

- Construction Contractor(s)
- Commissioning Contractor(s)
- Fabrication Contractor(s)
- Vendors
- Client(s)
- Engineering Contractor(s)

This functionality allows the relevant contractor/company to be allocated responsibility of punchlist items.

8.3.2.4 Punchlist Categories

In Brownfield projects punchlist items fall into two distinct categories. In the CMS each item is allocated either category A or B. The rules for which category an item should be allocated are as follows:

Category A

Items to be repaired, completed or corrected before the equipment can be safely commissioned and or operated. By definition all category A items shall be cleared prior to the livening up of equipment by Commissioning or Operations.

Category B

Items that may be repaired, completed or corrected at a later date which do not affect the safety of personnel, plant or environment whilst being commissioned or operated.

Under normal circumstances the Commissioning or Operations authorities will refuse to

accept ownership of any equipment if category A punchlist items are still outstanding.

8.3.3 Creating Punchlists

Punchlists are prepared and populated into the CMS with all other certification and handovers then printed and added to the workpack. As indicated above, it is good engineering practice to ensure that the scope of SH1s or MC1s are fully incorporated into a single construction workpack (i.e. SH1s and MC1s are never split across several construction workpacks).

A punchlist is required by discipline for each SH1, MC1 and IRN as per document *WCS-OPS-FLC-411 - Brownfield Functional Certification Flowchart*.

Punchlists are usually discipline based due to the fact that the tri-party walks are normally performed by discipline. This is not however always applicable or practical, particularly for smaller scopes of work whereby a multi-discipline inspection is a more sensible approach. This decision shall be made by the Construction Manager (for MC1 walks) or by the Commissioning Manager (for SH1 walks).

Once raised the punchlist form is then added to the relevant section of the construction workpack.

The punchlist shall be numbered as per following guidelines:

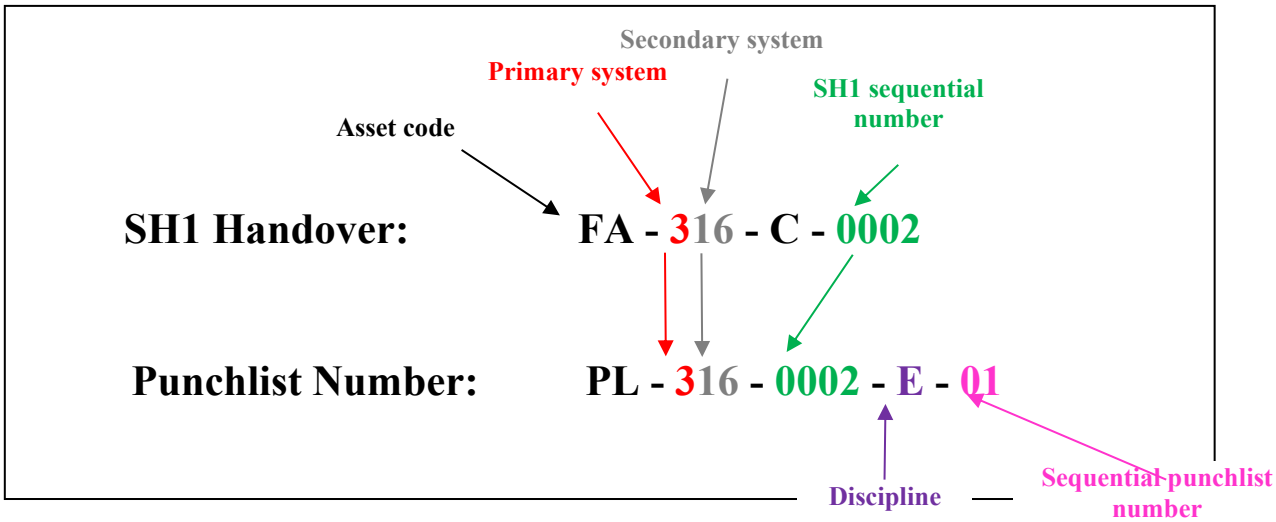

Figure 2 – Brownfield Punchlist Number Format for SH1s

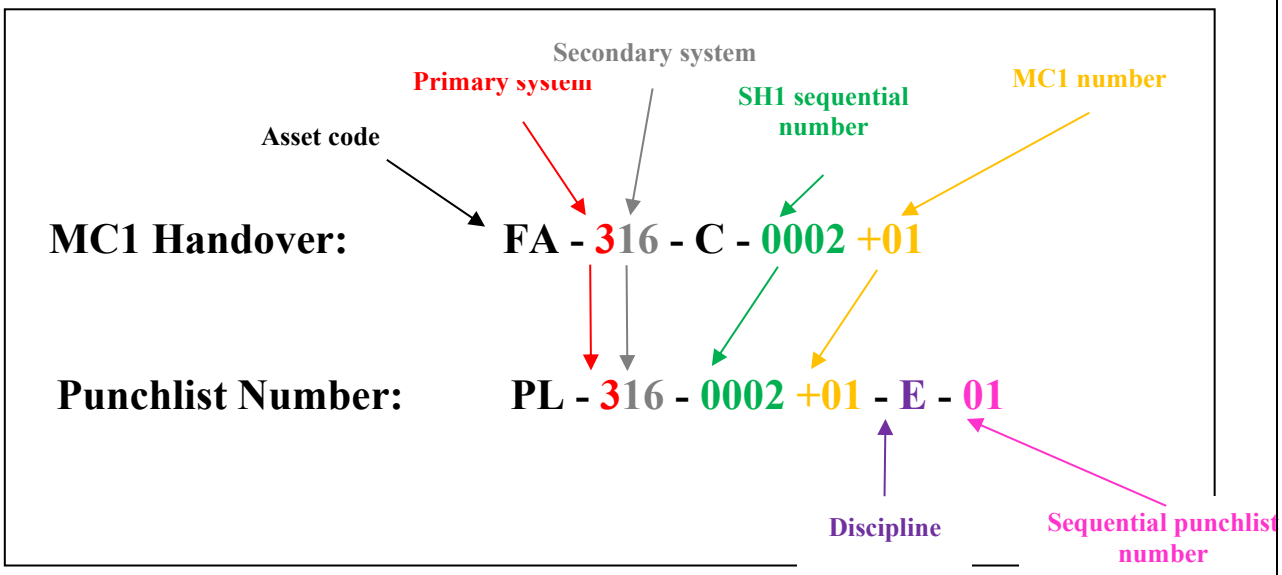

Figure 3 – Brownfield Punchlist Number Format for MC1s

IRN punchlists should use the MC1 punchlist number format.

8.3.3.1 Punchlist – Minimum Data

Punchlist Number – see guidelines above

Discipline

Description – Description for the punchlist stating the discipline and handover

Raised By – Person raising the punchlist

Date Raised – Date the punchlist was created

Location Originated – Location, or main location, where the workscope will be carried out

Type – The party who called for the punchlist inspection

SH1 – System handover subject to punchlist inspection

MC1 – Mechanical completion handover subject to punchlist inspection

8.3.3.2 Punchlist – Additional Data

Subsystem – For Brownfield the MC1 handovers are normally broken into subsystems for any given scope of work. In this instance subsystem is useful but is not a required field.

8.3.4 Raising Punchlists Items – Tri-Party Walkdowns

Punchlist items are identified during tri-party walkdowns. The walkdowns are usually performed by discipline for each IRN, MC1 and SH1, just before the handover is completed.

The tri-party walkdowns are arranged by the party handing over the scope of work. An invitation should be sent notifying all parties of the date and time of the walkdown. The notice period and list of competent representatives for each party involved in the handover process as well as the client shall be agreed at the start of the project.

Should a party representative be unable to attend the walkdown, a competent substitute must be nominated, and the other parties informed. Walkdowns shall not be cancelled or rescheduled due to the absence of a party.

On the day of the walkdown, the party calling the punchlist inspection shall provide a copy of the mark-ups showing the limits of system handover (LOSH) to be inspected for the MC1 and/or SH1 as well as the associated punchlist form(s) from the construction workpack.

The personnel raising punchlists items(s) should use the punchlist form provided in the construction workpack for the associated handover and discipline, completing as a minimum all mandatory data required for input by the Certification Engineer.

All punchlist items must be recorded on the form. At the end of the inspection the parties attending the walkdown shall agree the validity of the punchlist items, their categorisation and the party responsible for clearance. All representatives attending the walkdown shall sign the punchlist form, confirming their attendance and acknowledgment of all items listed.

Upon completion of the walkdown, in the event that no defects or non-conformance are identified, the text ‘NO PUNCHLIST ITEMS’ should be clearly marked on the punchlist form. All parties in attendance should sign the form to indicate their agreement prior to the handover taking place.

The punchlist form should then be returned to the Completions office, where it should be scanned, and an electronic copy emailed to the Certification Engineer in order for the CMS to be updated to reflect the current progress of the workscope.

8.3.5 Adding Punchlist Items and Punchlist Data Requirements

It is normally the responsibility of the Certification Engineer to input punchlist items into the CMS. When punchlist items have been added to the punchlist form it should be scanned and an electronic copy emailed to the Certification Engineer in order that the CMS can be updated to reflect the current progress of the workscope.

The Certification Engineer is responsible for requesting clarification from the relevant punchlist parties if the data provided on the punchlist form is insufficient.

8.3.5.1 Punchlist – Minimum Data

Punchlist Item Number - An incremental number

Description – An accurate description of the non-conformance or defect to be rectified and verified. Details should include, wherever possible, tag number and relevant drawings. A detailed description of the location should also be included in order that another person can find the equipment on site

Action Discipline – The discipline who will action the punchlist item

Category – Either A or B

Raised By – A person from the Project Signature Matrix with sufficient permission to raise punchlist items

Date Raised – The date the punchlist item was recorded

Action by - The designated party responsible for rectifying the non-conformance or defect (clearance body). In the CMS this is usually recorded in the 'Cleared By' and 'Cleared date' fields

Tag Number – The tag number of the defective equipment

8.3.5.2 Punchlist Item – Additional Data

ITR/Cert – If the punchlist item relates specifically to a certificate for any given equipment then the certificate should be included where possible

Location – Selected from the list of locations within the CMS

Manhours – An estimate of the manhours required to complete the punchlist item

Responsibility – The designated party responsible for verifying that the defect or non-conformance has been rectified (acceptance body). In the CMS this is usually recorded in the 'Accepted By' and 'Accepted date' fields

Materials – The MRO number, if material is on order

8.3.6 Managing the Punchlist

At the end of the tri-party walkdown the master copy of the punchlist form should be filed in the construction workpack as it will be used for clearing and accepting punchlist items. Should new punchlist items be raised after the tri-party walkdown, the party raising them is responsible for scanning and sending an electronic copy of the punchlist form, usually by email, to the Certification Engineer. The Certification Engineer should input new punchlist items into the CMS on a daily basis.

8.3.7 Punchlist Clearance and Acceptance

The Certification Engineer should export the outstanding punchlist items by MC1, discipline, area, and action by, for distribution to the parties involved in the punchlist clearance process. This should be as required or at a frequency to be determined by the project.

The punchlist parties will use this report to identify their scope of work. Under no circumstances should this report be used to signoff punchlist items. All clearance and acceptance must be recorded on the PL1 form.

Clearing and accepting punchlist items shall be carried out by competent personnel usually from the work group indicated by the Punchlist Party assignment.

Once the personnel in charge of clearing the punchlist item is satisfied that the defect has been rectified, they should locate the relevant item(s) on the punchlist form within the construction workpack and clear the item(s) by signing and dating the cleared by and cleared date fields.

The personnel responsible for accepting the punchlist item should verify that the defect or non-conformance has been rectified. Following verification, they should locate the relevant item(s) on the punchlist form within the construction workpack and accept the items by signing and dating the accepted by and accepted date fields.

As some signatures are not always easily identifiable it is good practice to add a full name, signature and date.

The punchlist form should then be scanned and an electronic copy emailed to the Certification Engineer in order that the clearance/acceptance signatures and dates can be input to the CMS. A person responsible for the scanning (if not performed by the Certification Engineer itself) should be identified prior to start of the punching activities.

8.3.8 Punchlist Reports

Two types of punchlist reports are typically required during the execution of the work:

- Punchlist status and progress report to project management:
  - Used to report the progress of punchlists items by SH1, MC1, discipline, category, etc to the Project Management team
  - Is not a list of punchlist items
  - Shows summary counts of punchlist items by SH1, MC1, disciplines, category and any other relevant breakdown
  - Shows outstanding and closed items
  - Shows a progress curve by discipline of items raised and closed over time
  - Punchlist report provided to OIM/FCO, by SH1, MC1 and discipline, for punchlist

clearance:

- Used to provide a list of outstanding items that require to be rectified/verified by the relevant punch list parties
- Lists all outstanding items relevant to the SH1, MC1, discipline, company and any other relevant filter
- Does not have sign off boxes
- Is sorted by SH1 and MC1 numbers

➢ Punchlist status report for inclusion in handover dossiers

- Used to report the outstanding and closed punchlist items at the time of the handover
- List of punchlist items relevant to the SH1/MC1
- Includes cleared by and accepted by names and dates

The Certification Engineer should run reports at an agreed frequency or as requested and send them to the OIM and or FCO to ensure they are apprised of progress and aware of the outstanding items.

8.3.9 Vendor Punchlist Requirement

All vendors must comply with the project punchlist procedure which should be issued through procurement.

Any issues found during FAT's must be recorded on a punchlist form. This should either be on the punchlist form associated with the handover or on the vendor equivalent. Vendor punchlist forms should only be used if the data requirement matches the project specification.

Upon completion of the FAT, the punchlist must be reviewed and its content and actionees approved by the client/EPC representative in attendance.

Once transmitted to the project this form must be passed to the Certification Engineer for input into the CMS. The copy of the vendor punchlist must be filed in the relevant construction workpack.

8.3.10 Punchlist Process Flow

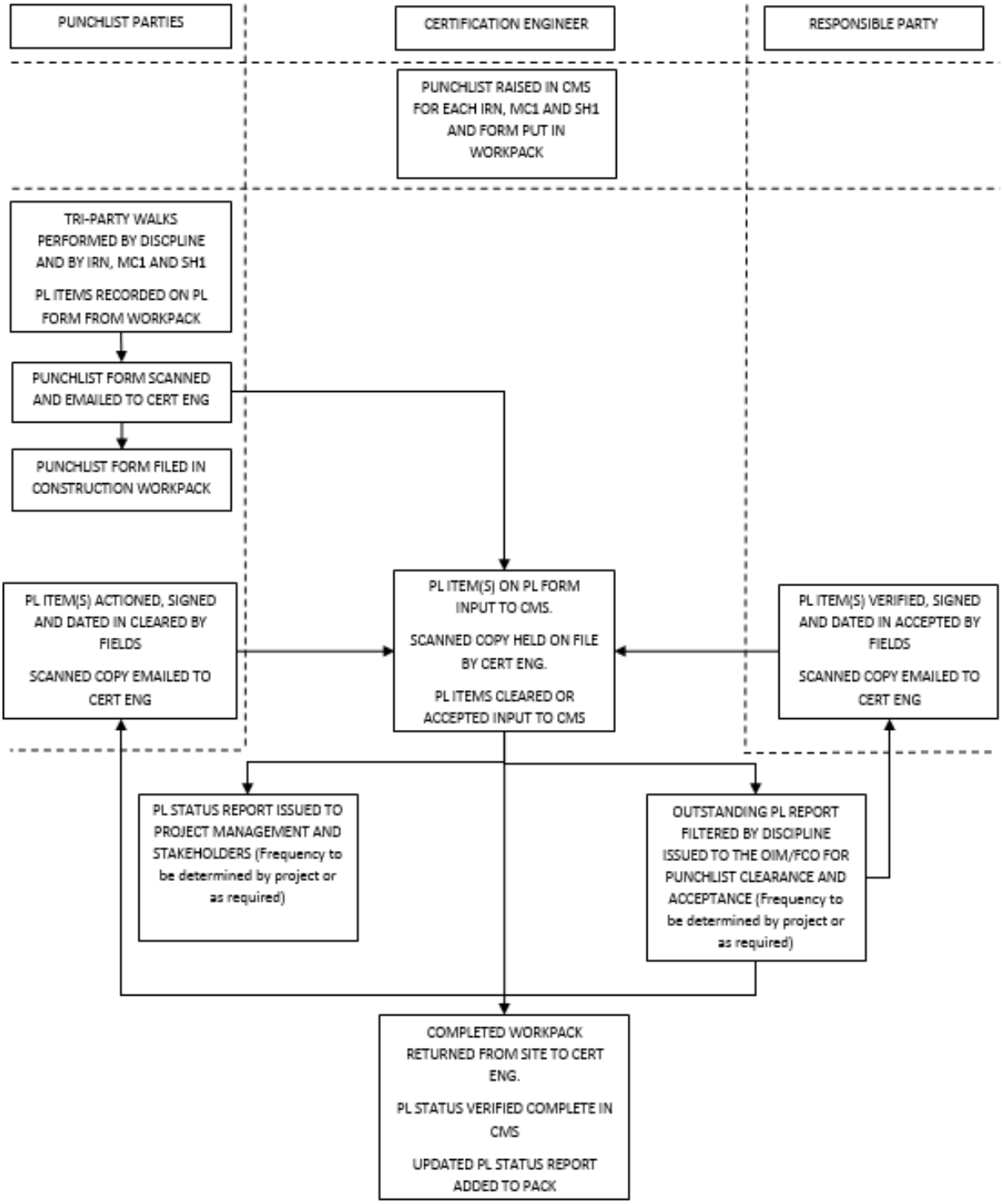

Figure 4 – Brownfield Punchlist Process Flowchart

APPENDIX-01 GREENFIELD & BROWNFIELD PUNCHLIST RAISE FORM